

From hyperlinks to Semantic Web properties using Open Knowledge Extraction

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Abstract. Open information extraction approaches are useful but insufficient alone for populating the Web with machine readable information as their results are not directly linkable to, and immediately reusable from, other Linked Data sources. This work proposes a novel Open Knowledge Extraction approach that performs unsupervised, open domain, and abstractive knowledge extraction from text for producing directly usable machine readable information. The method is based on the hypothesis that hyperlinks (either created by humans or knowledge extraction tools) provide a pragmatic trace of semantic relations between two entities, and that such semantic relations, their subjects and objects, can be revealed by processing their linguistic traces (i.e. the sentences that embed the hyperlinks) and formalised as Semantic Web triples and ontology axioms. Experimental evaluations conducted with the help of crowdsourcing confirm this hypothesis showing very high performances. A demo of Open Knowledge Extraction at <http://wit.istc.cnr.it/stlab-tools/legalo>.

Keywords: open knowledge extraction, open information extraction, abstractive summarisation, link semantics, relation extraction.

1. Introducing Open Knowledge Extraction

The vision of the Semantic Web is to populate the Web with machine understandable data so that intelligent agents are able to automatically interpret its content - just like humans do by inspecting Web content - and assist users in performing a significant number of tasks, relieving them of cognitive overload.

The Linked Data movement [2] kicked-off the vision by realising a key bootstrap in publishing machine understandable information mainly taken from structured data (typically databases) or semi-structured data (e.g. Wikipedia infoboxes). However, most of the

Web content consists of natural language text, hence a main challenge is to extract as much relevant knowledge as possible from this content, and publish them in the form of Semantic Web triples. This work aims at solving this problem by extracting relational knowledge that is “hidden” in hyperlinks, which can be either defined manually by humans (e.g. Wikipedia pagelinks) or created automatically by Knowledge Extraction (KE) systems (e.g. a KE system can automatically add links to Wikipedia pages).

Current knowledge extraction (KE) systems address very well the task of linking pieces of text to Semantic Web entities (e.g. `owl:sameAs`) by means

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of named entity linking methods, e.g. NERD¹ [43], FOX², conTEXT³ [26], Dbpedia Spotlight⁴, Stanbol⁵, TAGME [13], Babelfy [31]. Some of them (e.g. NERD) also perform sense tagging, i.e. adding knowledge about entity types (`rdf:type`).

Nevertheless, it is desirable to enrich Web content with other semantic relations than `owl:sameAs` and `rdf:type`, i.e. factual relations between entities. A *pragmatic trace* of a factual relation between two entities is the presence of a hyperlink, which is associated with its *linguistic trace*, i.e. the text surrounding the hyperlink. In fact, when we include a link in a Web page, we usually have a semantic relation in mind between something we are referring within the page, i.e. subject, and something referred by the target page, i.e. object, and the text where the hyperlink is embedded typically contains an explanation of what such relation is. For example, a link to “Usenet” in the Wikipedia page of “John McCarthy”⁶ suggests a semantic relation between those two entities, which is explained by the sentence: “McCarthy often commented on world affairs on the [Usenet forums](#)”.

Besides common sense, this hypothesis is also supported by a previous study [35], which describes the extraction of encyclopedic knowledge patterns for DBpedia types, based on links between Wikipedia pages. A user study showed that hyperlinks between Wikipedia pages determine relevant descriptive contexts for DBpedia entities at the type level, which suggests that these links mirror relevant semantic relations between entities.

A hyperlink in a Web page can be produced either by a human or a KE system (e.g., by linking a piece of text to a Wikipedia page, which in turn refers to a Semantic Web entity, i.e. a DBpedia entity). If a KE system recognises two or more entities in a sentence, there is a possibility that such sentence expresses some relation between them. For example, the following sentence:

The New York Times reported that John McCarthy died. He invented the programming language LISP.

can be automatically enriched using a KE system by linking the pieces of text “The New York Times”,

“John McCarthy”, and “LISP” to the Wikipedia pages `wikipedia:The_New_York_Times`⁷, `wikipedia:John_McCarthy_(computer_scientist)` and `wikipedia:Lisp_(programming_language)` (respectively), resulting in the following:

[The New York Times](#) reported that [John McCarthy](#) died. He invented the programming language [LISP](#).

In this example, the three hyperlinks identify entities that are relevantly related by factual relations: “John McCarthy” with “The New York Times”, and “John McCarthy” with “LISP”. Revealing the semantics of hyperlinks (either defined by humans or KE systems) has a high potential impact on the amount of Web knowledge that can be published in machine readable form.

In the Semantic Web era such factual relations should be expressed as RDF triples where subjects, objects, and predicates have a URI (except for literal objects and blank nodes), and predicates are formalised as RDF/OWL properties, in order to facilitate their reuse and alignment to existing vocabularies, and for example to annotate hyperlinks with RDFa, within HTML anchor tags.

While subjects and objects are mostly directly resolved on existing Semantic Web entities, predicates are to be defined by performing “paraphrasing”, a summarisation task able to abstract over the text (when needed) in order to design labels that are as close as possible to what a human would design in a Linked Data vocabulary. In this respect, [42] distinguishes between extractive and abstractive summarisation approaches. Extractive methods select pieces of texts from the original source in order to define a summary (i.e. they rely only on the available text), while abstractive techniques ideally rely on modeling the text, and then combining it with other resources and language generation techniques for generating a summary. Abstractive methods are usually applied to large documents to the aim of producing a meaningful summary of their content.

This work proposes to apply the guiding principle of abstractive techniques to open information extraction as a novel contribution. Open information extraction refers to an open domain and unsupervised extraction paradigm. Existing open information extraction approaches are mainly extractive, hence showing a complimentary nature to what we present in this

¹<http://nerd.eurecom.fr>

²<http://aksw.org/Projects/FOX.html>

³<http://context.aksw.org/app/>

⁴<http://dbpedia-spotlight.github.com/demo>

⁵<http://stanbol.apache.org>

⁶Cf. [http://en.wikipedia.org/wiki/John_McCarthy_\(computer_scientist\)](http://en.wikipedia.org/wiki/John_McCarthy_(computer_scientist))

⁷wikipedia: stands for <http://it.wikipedia.org/wiki/>

Subject	Predicate	Object	Approach
John Stigall	received	a Bachelor of arts	extractive
John Stigall	received	from the State University of New York at Cortland	extractive
dbpedia:John_Stigall	myprop:receive_academic_degree	dbpedia:Bachelor_of_arts	abstractive
dbpedia:John_Stigall	myprop:receive_academic_degree_from	dbpedia:State_University_of_New_York	abstractive

Table 1

Comparison between relations resulting from extractive and abstractive approaches for the sentence “[John Stigall](#) received a [Bachelor of arts](#) from the [State University of New York at Cortland](#)”.

paper. They mostly focus on breaking text in meaningful pieces for building resources of relational patterns (e.g. PATTY [32]⁸, Wisenet [30]⁹), in some cases disambiguated on external semantic resources such as WordNet¹⁰. Others focus on extracting facts, which are represented as simplified strings between entities (e.g. Open Information Extraction (OIE) [28]¹¹) that are not given a Semantic Web identity.

Knowledge extraction for the Semantic Web should instead include an abstractive step, which exploits a formal semantic representation of text, and produces output that is compliant with Semantic Web principles and requirements. The method described in this paper demonstrates a novel unsupervised, open domain and abstractive paradigm, driven by Semantic Web principles and requirements, called *open knowledge extraction* (OKE). For example, given the sentence:

[John Stigall](#) received a [Bachelor of arts](#) from the [State University of New York at Cortland](#).

Table 1 compares the extracted relations resulting from an extractive approach (such as OIE [28]¹²) - the first two rows - and from an abstractive approach - the last two rows. The abstractive results exemplify the expected result of a OKE method. The main difference is that with the abstractive approach, subjects and objects are identified as Semantic Web entities, the predicate is as close as possible to what a human would define for a Linked Data vocabulary by possibly using terms that are not mentioned in the original text. In addition to what Table 1 shows, the predicate would be formally defined in terms of OWL axioms and possibly aligned with existing Semantic Web vocabularies.

1.1. Contribution

Open information extraction approaches are useful but insufficient alone for populating the Web with machine readable information as their results are not directly linkable to, and immediately reusable from, other Linked Data sources. This work main contributions are:

- a novel approach *Open Knowledge Extraction* (OKE) approach that performs unsupervised, open domain, and abstractive knowledge extraction from text for producing directly usable machine readable information;
- an implementation of OKE, named *Legalo*¹³ that given an English sentence produces a set of RDF triples representing relevant factual relations expressed in the sentence, the predicates of which are formally defined in terms of OWL axioms;
- an evaluation of Legalo performed on a number of sample sentences taken from a corpus¹⁴ of validated sentences that provide evidence of factual relations. The results have been evaluated with the help of crowdsourcing and the creation of a gold standard, all showing high values of precision, recall, and accuracy;
- a discussion highlighting the current limits of the approach and possible ways of improving it, and including an informal comparison of the proposed method with one of the main existing open information extraction tools.

Additionally, the paper includes a brief description of a specific implementation of OKE, specialised for extracting the semantics of Wikipedia pagelinks, which has been evaluated in [40] showing promising results.

⁸<https://d5gate.ag5.mpi-sb.mpg.de/pattyweb/>

⁹<http://lcl.uniroma1.it/wisenet/>

¹⁰<http://wordnet.princeton.edu/>

¹¹<http://openie.cs.washington.edu/>

¹²Notice that this is the output of OIE for this sentence

¹³Legalo demo can be played at <http://wit/istc.cnr.it/stlab-tools/legalo/>

¹⁴<https://code.google.com/p/relation-extraction-corpus/>

The paper is structured as follows: Section 2 presents the novel OKE method. Section 3 describes data sources that were used for developing a demonstrator of OKE and for its evaluation, while Section 4 describes *Legalo*, an implementation of OKE that has been evaluated with the help of crowdsourcing, as described in Section 5. Section 6 discusses evaluation results, limits of the method and possible ways to improve it, and informally compares Legalo with Open Information Extraction (OIE) [28]. Section 7 discusses relevant research work and finally, Section 8 summarises the contribution of this work and briefly indicates future works.

2. A method for generating Semantic Web properties from text

OKE addresses the following capabilities:

- to assess if a natural language sentence provides an evidence of a relevant relation between a given pair of entities, which are identified by hyperlinks; *relevant* here means that there are enough explicit traces in the sentence to support the existence of a (conceptual) relation;
- to generate a predicate for this relation, with a label that is as close as possible to what a human would define in a Linked Data vocabulary;
- to formalise this relation as an OWL object property with TBox axioms (conceptual level), as well as to produce ABox axioms (factual level) using that property.

More formally:

Definition 1. (Relevant relation)

Let s be a natural language textual sentence embedding some hyperlinks, and (e_{subj}, e_{obj}) a pair of entities mentioned in s , where e_{subj} and e_{obj} are the target entities referred by two hyperlinks in s , $\varphi_s(e_{subj}, e_{obj})$ is a relevant relation between e_{subj} and e_{obj} , expressed in s , with e_{subj} being the subject of φ and e_{obj} being its object. $\Lambda \equiv \{\lambda_1, \dots, \lambda_n\}$ is a set of Linked Data labels generated by humans for $\varphi_s(e_{subj}, e_{obj})$. Finally, λ' is a label generated by OKE for $\varphi_s(e_{subj}, e_{obj})$.

OKE is able to assess the existence of $\varphi_s(e_{subj}, e_{obj})$ and to generate a label λ' equal or very similar to $\lambda_i \in \Lambda$. OKE method is based on six main steps:

1. (*abstractive step*) internal formal representation of the sentence;
2. assessment of the existence of a relevant relation between pairs of entities identified in s , according to the content of sentence;
3. (*extractive step*) extraction of relevant terms for the predicate;
4. (*abstractive step*) generation of the predicate label;
5. (*abstractive step*) formal definition of the predicate within the scope of its linguistic evidence and formal representation;
6. alignment (whenever possible) to existing Semantic Web properties.

2.1. Frame-based formal representation of a sentence

Given the capabilities described in Section 2, i.e. the assessment of $\varphi_s(e_{subj}, e_{obj})$ existence and the generation of λ' (cf. Definition 1), for both tasks OKE relies on a set of rules to be applied to a frame-based formal representation G of the sentence s (cf. Definition 2). G is a RDF graph designed following a frame-based approach, where nodes represent entities mentioned in s .

Definition 2. (Frame-based graph)

Let s be a natural language text sentence and $G = (V, E)$ a RDF (directed, multi-) graph modelling a frame-based formal representation of s , where $V \equiv \{v_0, \dots, v_n\}$ is the set of nodes (i.e. subjects and objects from RDF triples) in G , $E \equiv \{edge_1, \dots, edge_n\}$ is the set of edges (i.e. RDF triples) in G , where $edge_i = (v_{i-1}, p, v_i)$, is a triple connecting v_{i-1} and v_i with the RDF property p in G , and $v_i \in V$ is the node in G representing the entity e_i mentioned in s .

Frame Semantics [14] is a formal theory of meaning: its basic idea is that humans can better understand the meaning of a single word by knowing the relational knowledge associated to that word. For example, the sense of the word *buy* can be clarified in a certain context or task by knowing about the situation of a commercial transfer that involves certain individuals playing specific roles, e.g. a seller, a buyer, goods, money, etc.

In this work, frames are usually expressed by verbs or other linguistic constructions, and their occurrences in a sentence are represented as RDF n -ary relations, all being instances of some type of event or situation (e.g. `myont:buy_1 rdf:type myont:Buy`), which is on its turn represented as a subclass of

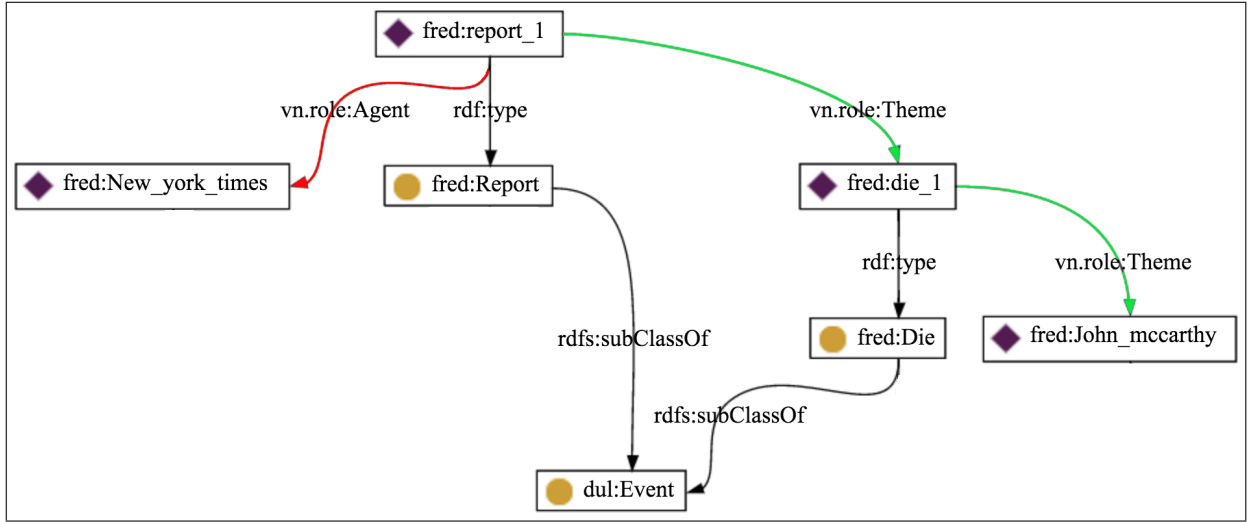


Fig. 1. Frame-based formal representation for the sentence: “The New York Times reported that John McCarthy died.”

`dul:Event`¹⁵. Intuitively, `dul:Event` is the top category of all frames expressed by verbs. In the context of this paper, the terms frame occurrence and event occurrences are used as synonyms. Entities that are mentioned in s are represented as individuals or classes, depending on their nature, which (ideally) have a type, defined based on the information available in the sentence s . When appropriate, entities are represented as arguments of n -ary relations, according to the role they play in the corresponding frame occurrence. The role of an entity in an event occurrence can be expressed either by a preposition, e.g. *Rico Lebrun taught at the Chouinard Art Institute*, or it can be abstracted from the text and represented by reusing the set of thematic roles defined by VerbNet [44], e.g. Rico Lebrun is the **agent** of the event occurrence “teach” in the above sample sentence.

A formal and detailed discussion of the theory behind frame-based formal representation of knowledge extracted from text, and used by OKE is beyond the scope of this paper. This modeling approach and its founding theories are extensively described in [14,41,34]. However an example may be useful to convey the intuition behind the theory. Figure 1 shows a frame-based representation of the sentence:

[The New York Times](#) reported that [John McCarthy](#) died.

The knowledge extracted from the sentence s is formalised as a set of RDF triples G ¹⁶. Two entities can be identified in this sentence, i.e. “New York Times” and “John McCarthy”, represented in G as individuals i.e. `fred:New_York_Times` and `fred:John_McCarthy`, respectively. Two frame occurrences can be identified in the sentence: one expressed by (an inflected form of) the verb *report* and the other expressed by the verb *die*. These frame occurrences are represented as n -ary relations: i.e., `fred:report_1` and `fred:die_1`, both being instances of classes (`fred:Report` and `fred:Die` respectively) that are of type `dul:Event`. Let us consider the event occurrence `fred:report_1`. Its arguments are: (i) `fred:New_York_Times`, which plays an *agentive role* in this event occurrence, formally expressed by the predicate `vn.role:Agent`¹⁷, and `fred:John_McCarthy`, who plays a *passive role*, formalised by the predicate `vn.role:Theme`, both VerbNet thematic roles.

2.2. Relevant relation assessment

OKE method for assessing if a relevant relation $\varphi_s(e_{subj}, e_{obj})$ exists in s between a pair of entities (e_{subj}, e_{obj}) relies

¹⁵The prefix `dul:` stands for <http://www.ontologydesignpatterns.org/ont/dul/dul.owl#>

¹⁶The figure is derived from the output of FRED [41] (see Section 4), the component providing the frame-based formal representation within OKE implementation. The prefix `fred:` stands for a local configurable namespace.

¹⁷Prefix `vn.role:` stands for <http://www.ontologydesignpatterns.org/ont/vn/abox/role/>, which defines all VerbNet [44] thematic roles.

on the analysis of the semantic structure of G . Firstly, $\varphi_s(e_{subj}, e_{obj})$ is assumed to hold only if there is at least one path in G connecting v_{subj} and v_{obj} , i.e. the nodes representing e_{subj} and e_{obj} in G , regardless of the edge direction in G . This is formally expressed by Axiom 1, given Definition 3.

Definition 3. (Graph path)

$G' = (V, E')$ is the undirected version of $G = (V, E)$. A path $P(v_{subj}, v_{obj}) = [v_0, edge_1, \dots, edge_n, v_n]$ with $v_0 = v_{subj}$ and $v_n = v_{obj}$ is any sequence alternating nodes and edges in G' connecting v_{subj} to v_{obj} , or vice versa. The set $Pset_{subj,obj} \equiv \{edge_1, v_1, \dots, edge_n\}$ includes all edges and nodes in $P(v_{subj}, v_{obj})$ excluding v_{subj} and v_{obj} .

Axiom 1. (φ assessment: necessary condition)

$$\varphi_s(e_{subj}, e_{obj}) \Rightarrow \exists P(v_{subj}, v_{obj})$$

If $P(v_{subj}, v_{obj})$ exists, OKE distinguishes whether $P(v_{subj}, v_{obj})$ contains an event occurrence, or not. If $P(v_{subj}, v_{obj})$ does not contain any event occurrence, then the existence of $P(v_{subj}, v_{obj})$ is a sufficient condition to the existence of $\varphi_s(e_{subj}, e_{obj})$ (cf. Axiom 2).

Axiom 2. (Assessment of φ : sufficient condition without event occurrences)

$$\varphi_s(e_{subj}, e_{obj}) \Leftarrow \exists P(v_{subj}, v_{obj}) \text{ such that } \forall v_i \in Pset_{subj,obj}, \neg \text{dul} : \text{Event}(v_i)$$

In the other case, i.e. the path includes an event occurrence, OKE method states that $\varphi_s(e_{subj}, e_{obj})$ exists if e_j is the subject of the event verb in the sentence. In the graph G this means that the node v_{subj} representing e_{subj} in G participates in the event occurrence with an agentive role. This is formalised by Axiom 3, given Definition 4.

Definition 4. (Agentive roles)

Let f be a node of G such that $\text{dul} : \text{Event}(f)$ $\text{Role} \equiv \{\rho_1, \dots, \rho_n\}$ is the set of possible roles participating in f , $\text{AgRole} \equiv \{\rho_{m_1}, \dots, \rho_{m_m}\}$ is the set of VerbNet agentive roles, with $\text{AgRole} \subseteq \text{Role}$, and $\rho(f, v_{subj})$ is a role connecting the event occurrence f to its participant v_{subj} (the node representing e_{subj}) in s .

Axiom 3. (Assessment of φ with event occurrences: sufficient condition)

$$\varphi_s(e_{subj}, e_{obj}) \Leftarrow \exists P(v_{subj}, v_{obj}) \text{ and } \exists f \in Pset_{subj,obj} \text{ such that } \text{dul} : \text{Event}(f) \text{ and } \rho(f, v_{subj}) \in \text{AgRole}$$

This axiom is based on linguistic typology results (e.g. [8]), by which SVO (Subject-Verb-Object) languages such as English have almost always an explicit (or explicitatable) subject. This subject is formalized in a frame-based representation of s by means of an agentive role. Based on this observation, OKE method states that $\varphi_s(e_{subj}, e_{obj})$ exists if e_{subj} is the subject of a verb in s . This axiom is potentially restrictive with respect to the idea of a relevant relation expressed in a sentence, which may consider any pair of entities as related just because they are mentioned in a same sentence. In fact, this idea is quite difficult to implement, since relations between pairs of entities that play e.g. oblique roles¹⁸ in a frame occurrence are hard to paraphrase even for a human. For example, consider the sentence:

After a move to [Southern California](#) in 1938, [Rico Lebrun](#) taught at the [Chouinard Art Institute](#) and then at the [Disney Studios](#).

the frame-based representation of this sentence, depicted in Figure 2 identifies Rico Lebrun as the agent of a “teach” frame occurrence, while Southern California, Chouinard Art Institute, and Disney Studios participate in it with oblique roles. This sentence expresses three relevant relations: one between Rico Lebrun and Chouinard Art Institute, one between Rico Lebrun and Disney Studios, and another between Rico Lebrun and Southern California. All those relations can be easily summarized and represented as RDF triples, typically selected by OKE from the knowledge extracted from this sentence.

On the other hand, while it is correct to state that Chouinard Art Institute and Disney Studios co-participate in an occurrence of the frame “teach”, it is far from straightforward to paraphrase the meaning of this relation. E.g., one might say that Chouinard Art Institute and Disney Studios are both places where Rico Lebrun used to teach, but this paraphrase is not easily reconstructable from the text, and needs a stronger language generation approach, which has not been tackled for the moment. OKE may still represent this relation inferentially as a generic co-participation relation, which is however too generic to be considered as relevant.

For this reason, the investigation of paraphrases of relation between entities co-participating in an event

¹⁸Oblique roles are neither agentive or passive, e.g. “manner”, “location”, etc.

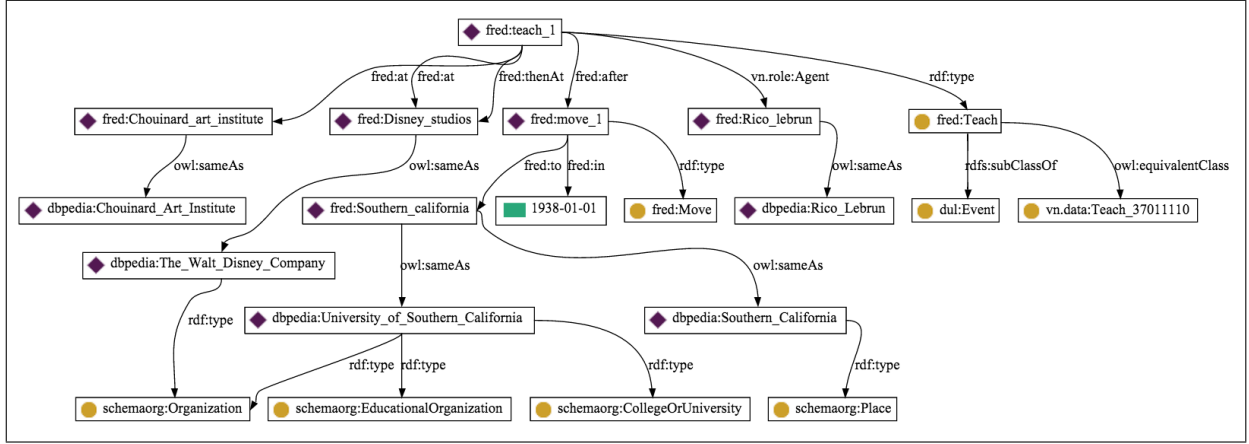


Fig. 2. Frame-based formal representation for the sentence: “After a move to [Southern California](#) in 1938, [Rico Lebrun](#) taught at the [Chouinard Art Institute](#) and then at the [Disney Studios](#)”. OKE will select the pairs of entities (fred:Rico_lebrun, Chouinard_art_institute), (fred:Rico_lebrun, fred:Disney_studios), and (fred:Chouinard_art_institute, fred:Southern_California)

with oblique roles is left to further study. An interesting analysis on this problem that could suggest new work directions is discussed in [9].

2.3. Combining extractive and abstractive design for property label generation

As far as the generation of λ' is concerned (cf. Definition 1), OKE combines extractive with abstractive techniques [42]. It means that it both reuses the terms in the text (extractive) and generates other terms derived from a semantic analysis of the text (abstractive). To this aim, OKE uses the semantic information provided by the frame-based representation G of the sentence s , which is further enriched with knowledge retrieved from external semantic resources. The resources DBpedia [3], Schema.org¹⁹, WiBi [15], and VerbNet [44] are used as examples as they are included in the OKE implementation presented in this work, i.e. Legalo (cf. Section 4). In particular, DBpedia is used for resolving (disambiguating) the nodes $\{v_i\} \in V$ that represent the entities $\{e_i\}$ in the sentence s , on Linked Data. WiBi and Schema.org are used for retrieving the types of these entities, and the labels of WiBi types are used in the generation of λ' . From VerbNet, as anticipated in Section 2.1, OKE obtains the thematic roles played by the DBpedia entities participating in a frame occurrence. Additionally, VerbNet is also used for disambiguating the sense of frame occurrences. For example, consider the sentence:

In February 2009 [Evile](#) began the pre-production process for their second album with [Russ Russell](#).

Figure 3 shows the enriched frame-based formal representation of this sentence. The graph does not show WiBi types but they are actually retrieved by OKE implementation, for each resolved DBpedia entity. Two entities are resolved on DBpedia, i.e. dbpedia:Evil, and dbpedia:Russ_Russell, and two frame occurrences are identified, i.e. fred:begin_1 and fred:process_1. Furthermore, each node is assigned with a type that, when possible, is aligned to existing Linked Data vocabularies. For Example, dbpedia:Evil has type schema.org:MusicGroup²⁰, and the entity fred:album_1 (representing the album mentioned in the sentence) is typed by the taxonomy fred:SecondAlbum rdf:type fred:Album. Following Axiom 1 and Axiom 3 (cf. Section 2.2), OKE will select from the graph of Figure 3 the pair of (DBpedia) entities:

dbpedia:Evil, dbpedia:Russ_Russell

OKE design strategy for generating predicate labels is based on three main generative rules (GR). The first one concerns the concatenation of the labels that are used in the shortest path connecting the two nodes, including the labels of the edges and the labels of the node types in the path. This rule is defined by GR 1. It is important to remark that the path used as a reference for generating the predicate label is the one connect-

¹⁹<http://schema.org/>

²⁰Prefix schema.org: stands for <http://schema.org>

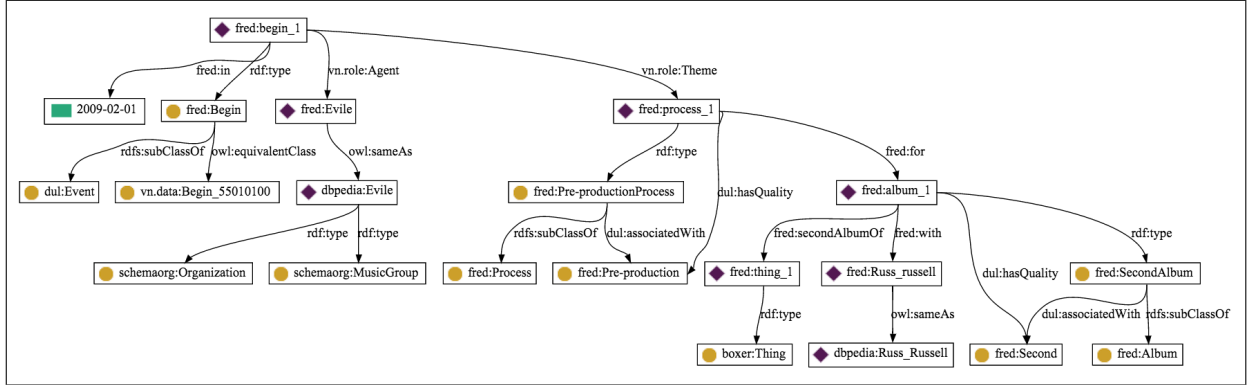


Fig. 3. Frame-based formal representation for the sentence: “In February 2009 [Evile](#) began the pre-production process for their second album with [Russ Russell](#)” The graph is enriched with verb senses to disambiguate frame types, DBpedia entity resolutions, thematic roles played by DBpedia entities participating in frame occurrences, and entity types.

ing the nodes v_{subj} and v_{obj} and not the corresponding resolved DBpedia entities.

GR 1. (Labels concatenation)

Given a pair (v_{subj}, v_{obj}) :

- identify the shortest path(s) $P(v_{subj}, v_{obj})$ connecting v_{subj} and v_{obj} ;
- extract all labels (matching sentence terms) of the edges in the path;
- extract all labels of the most general types²¹ of the nodes that compose the path, except the types of v_{subj} and v_{obj} ;
- concatenate the extracted labels following their alternating sequence in $P(v_{subj}, v_{obj})$.

Hence, referring to Figure 3, OKE will produce a predicate label λ = “begin process for album with” for expressing $\varphi_s(Evile, Russ\ Russell)$. Notice that the only labels that are included in the concatenation are those with prefix `fred:` meaning that they are extracted from s .

The second rule for generating predicate labels takes into account the possible presence of an event occurrence in the path connecting the pair (v_{subj}, v_{obj}) . Intuitively, in this case the path is a tree, rooted in an event occurrence, i.e. a node f , such as `dul:Event(f)`. The labels in this cases are extracted only from the path starting from f and ending in v_{obj} (referred as the right branch of the tree), including also the label of f type. The rationale behind this rule is that the right branch of the tree including the root event (i.e. its type) provides the relevant information expressing the rela-

tion between the two nodes, according to an empirical observation conducted on a sample of ~ 200 cases.

For example, consider the (excerpt of the) frame-based representation of the sentence “[Joey Foster Ellis](#) has published on [The New York Times](#), and [The Wall Street Journal](#).” shown in Example 2.1

Example 2.1. (Path including an event)

```
fred:publish_1 rdf:type fred:Publish;
  vn.role:Agent fred:Joey_Foster_Ellis;
  fred:on fred:New_York_Times;
  fred:on fred:Wall_Street_Journal .
fred:Publish rdfs:subClassOf dul:Event .
fred:Joey_Foster_Ellis
  owl:sameAs dbpedia:Joey_Foster_Ellis .
fred:New_York_Times
  owl:sameAs dbpedia:The_New_York_Times .
fred:Wall_Street_Journal
  owl:sameAs dbpedia:Wall_Street_Journal.
```

Following GR 1 and applying this additional rule for the selected pair:

```
dbpedia:Joey_Foster_Ellis,
```

```
dbpedia:Wall_Street_Journal
```

leads to a predicate λ = “publish on” for

$\varphi_s(Joey\ Foster\ Ellis, Wall\ Street\ Journal)$.

Additionally, if the right branch of the tree path is of length 1 and the only edge is a passive role, i.e. v_{obj} participates with a passive role to f , the label of the WiBi type of v_{obj} is concatenated to the predicate label. The rationale behind this rule is that when v_{subj} and v_{obj} play respectively an agentive and a passive role in an event occurrence, the resulting predicate label following only GR 1 would be too generic, hence

²¹If a node is typed by a taxonomy, the most general type in the taxonomy is extracted.

adding the WiBi type label makes the property label more specific and informative.

For example, a frame-based representation of the sentence “[Elton John](#) plays the [piano](#)” is given in Example 2.2:

Example 2.2. (Right branch of tree path with only passive role)

```
fred:play_1 rdf:type fred:Play;
  vn.role:Agent fred:Elton_John;
  vn.role:Theme fred:piano_1 .
red:Elton_John
  owl:sameAs dbpedia:Elton_John .
fred:piano_1 rdf:type dbpedia:Piano .
fred:Play rdfs:subClassOf dul:Event .
dbpedia:Piano
  rdf:type wibi:MusicalInstrument .
```

If we apply the additional rules described so far to the pair (dbpedia:Elton_John, dbpedia:Piano) we obtain a label λ = “play musical instrument” for $\varphi_s(\text{Elton John}, \text{piano})$, which is more informative than a simple “play” that would result without adding the WiBi type label of dbpedia:Piano. This rule is defined by GR 2.

GR 2. (Path including event occurrences)

Given a selected pair (v_{subj}, v_{obj}) and the shortest path $P(v_{subj}, v_{obj})$ connecting them. If $P(v_{subj}, v_{obj})$ is a tree rooted in f , such as $dul:Event(f)$,

- extract $P(f, v_{obj})$ from $P(v_{subj}, v_{obj})$;
- extract all edge labels in $P(v_{subj}, v_{obj})$ that match with terms extracted from s ;
- for each v_i (including f and excluding v_{obj}) in $P(f, v_{obj})$ extract the label of its more general type;
- concatenate the extracted labels following their alternating sequence in $P(f, v_{obj})$;
- if $P(f, v_{obj})$ has only 1 edge (length = 1), and this edge identifies a VerbNet passive role, then extract the WiBi type of v_{obj} and append it to the label concatenation.

The third rule for predicate label generation complements GR 1 and GR 2 by associating VerbNet roles to labels. Such labels have been defined top-down by analysing VerbNet thematic roles and their usage examples. The rule is defined in GR 3.

GR 3. (Thematic roles labels)

If a path contains a VerbNet thematic role, replace its label with an empty one, unless the role is associ-

ated with a non empty label according to the following scheme:

```
vn.role:Actor1 -> “with”
vn.role:Actor2 -> “with”
vn.role:Beneficiary -> “for”
vn.role:Instrument -> “with”
vn.role:Destination -> “to”
vn.role:Topic -> “about”
vn.role:Source -> “from”
```

For example, consider the (excerpt of the) frame-based representation of the sentence “[Lincoln’s wife](#) suspects that [John Wilkes Booth](#) and [Andrew Johnson](#) conspired to kill [Lincoln](#).” shown in Example 2.3.

Example 2.3. (Thematic roles associated with labels)

```
fred:conspire_1 rdf:type fred:Conspire;
  vn.role:Actor1 fred:Andrew_Johnson;
  vn.role:Actor2 fred:John_Wilkes_Booth .
fred:Conspire rdfs:subClassOf dul:Event .
fred:Andrew_Johnson
  owl:sameAs dbpedia:Andrew_Johnson .
fred:John_Wilkes_Booth
  owl:sameAs dbpedia:John_Wilkes_Booth;
fred:Lincoln
  owl:sameAs dbpedia:Abraham_Lincoln .
```

By applying GR 1, 2 and 3 to the path connecting the pair:

```
dbpedia:Andrew_Johnson,
dbpedia:John_Wilkes_Booth
```

OKE generates a label λ = “conspire with” for $\varphi_s(\text{Andrew Johnson}, \text{John Wilkes Booth})$.

The mapping scheme (role $\leftrightarrow$ label) is an evolving resource, which improves based on the periodic evaluation of OKE implementation results.

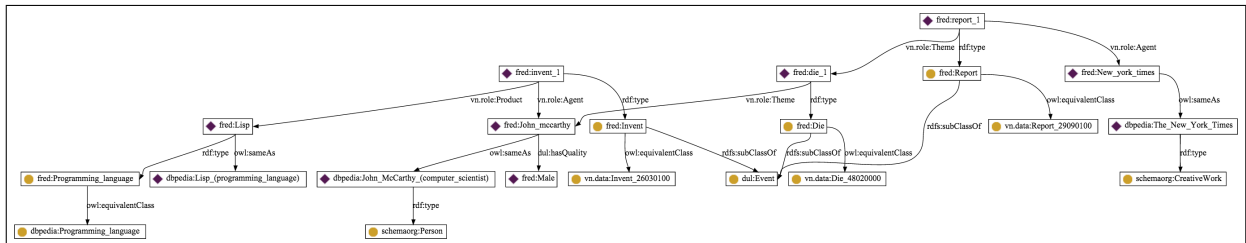
2.4. Formalisation of extracted knowledge

Given a textual sentence s and its frame-based formal representation G , by following the generative rules described in Section 2.3 Legalo generates a label λ for each relation $\varphi_s(e_{subj}, e_{obj})$ that it is able to identify in s , based on the shortest path $P(v_{subj}, v_{obj})$ connecting (v_{subj}, v_{obj}) in G (cf. Definitions 1, 2, and 3). These labels constitute the basis for automatically generating a set of RDF triples that can be used for semantically annotating the hyperlinks included in s , additionally these set of triples provides a (formalised) summary of s .

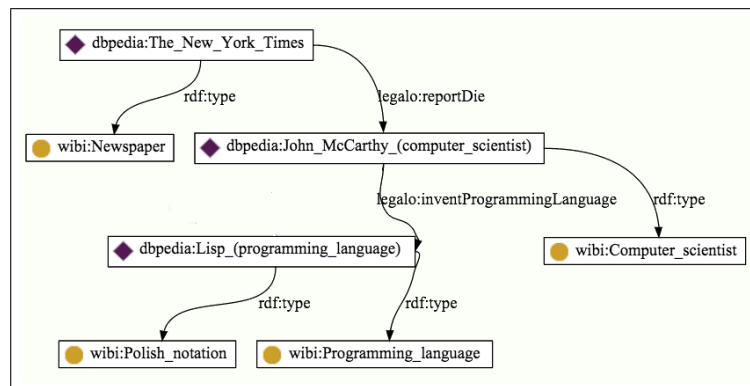
reuse, and scope annotations (i.e. linguistic evidence and formal representation) support reuse in relation extraction systems, e.g. relation extraction based on distant supervision [29,1]

- Locality of produced predicates.* OKE method works on the assumption that each generated predicate and its associated formalisation are valid in the conceptual scope identified by the sentence s . This means that s identifies the scope of predicate names definitions, i.e. the namespace of a predicate depends on s . Pragmatically, this is implemented in Legalo by including the checksum of s in the predicate namespace. This strong locality constraint may lead to producing a high number of potentially equivalent properties (i.e. having the same intensional meaning) defined as they were different. This issue is tackled by formalising all predicates with domain and range axioms having values, i.e. classes, from external (open domain) resources, as well as by keeping the binding between a predicate, its linguistic evidence, i.e. s , and its formal representation source, i.e. G . The latter containing information about the disambiguated senses of the verbs, i.e. frame occurrences, used in s . All these features allow on one hand to inspect a specific property for under-

on one hand to inspect a specific property for under-



ming language LISP.”



language LISP.”

standing its meaning, e.g. in case of manual reuse, on the other to automatically reconcile predicates by computing a similarity measure based on them. In this paper, we focus on the generative part of the problem, i.e. generating usable labels for predicates and producing their formal definition, while we leave the reconciliation task to future work.

RDF factual statements. For each hyperlink in s associated with a true assessment of $\varphi_s(e_{subj}, e_{obj})$ (cf. Axioms 1, 2, and 3), OKE produces at least one RDF triple. As explained in Section 2.3, the nodes v_{subj} and v_{obj} in G representing e_{subj} and e_{obj} are resolved on DBpedia, which ensures that all produced triples are linked to the Linked Data cloud. The predicate is formalised as an OWL object property having λ' as label and an ID derived by transforming λ' according to the CamelCase notation²².

For example, consider the enriched frame-based formal representation of the sentence

[The New York Times](#) reported that [John McCarthy](#) died. He invented the programming language [LISP](#).

depicted in Figure 4, OKE (i.e. its implementation) produces the triples depicted in Figure 5, according to the generative rules GR 1, 2, and 3, where the prefix `legalo:` is a namespace defined using the checksum of the sentence s . Notice that Figure 5 shows the WiBi types²³ for the resolved DBpedia entities.

OWL property formalisation. For each generated property, OKE produces an additional set of OWL axioms that formally define it. The predicate formalisation states that the predicate is an OWL object property, and includes domain and range axioms, whose values are defined according to the WiBi types assigned to v_{subj} and v_{obj} . In case of multi-typing of an entity, the value is the union of all types. In case a WiBi type is not available, the default type is `owl:Thing`. Example 2.4 shows the axioms formalising domain and range of the properties shown in Figure 5.

Example 2.4. (Domain and range axioms.)

```
legalo:reportDie a owl:ObjectProperty ;
  rdfs:domain wibi:Newspaper ;
  rdfs:range wibi:Computer_scientist .
```

²²According to a common Linked Data convention, using the CamelCase notation for OWL object properties makes the first term of the ID start with lower case, e.g. “invent programming language” -> `inventProgrammingLanguage`.

²³<http://www.wibitaxonomy.org/>

```
legalo:inventProgrammingLanguage
  a owl:ObjectProperty ;
  rdfs:domain wibi:Computer_scientist ;
  rdfs:range wibi:Programming_language ;
  rdfs:subPropertyOf legalo:invent .
```

As the reader may notice, an additional `rdfs:subPropertyOf` axiom is included in the formal definition of `legalo:inventProgrammingLanguage`. In fact, if a predicate is derived with GR 2, meaning that v_{subj} and v_{obj} participate in an event with respectively, an agentive and a passive role, then OKE also generates a more general property based on the event type, and produces a `rdfs:subPropertyOf` axiom. We remind that in these cases, the rule requires to generate a specialised property label by appending the WiBi type of v_{obj} to the label of the event type. Example 2.4 shows one of this cases. All properties produced by OKE are derived from a formal representation G of the sentence s , meaning that G provides their formal scope. Based on this principle, OKE produces an additional set of triples, which formalise the generated properties with reference to G . As stated by GR 1 and 2, there are two main types of paths from which the properties derive. In the first case, the path connecting v_{subj} and v_{obj} does not include any event node. In this case, OKE produces a OWL property chain axiom stating that the generated property is implied by the chain of properties participating in the path, where each property of the path is formalised with domain and range axioms according to the locality of G . The same concept applies to the case of a path that includes an event node. Similarly, OKE produces a property chain axiom. However, in this case the path has two different directions in G . For this types of paths OKE defines the concepts of *left branch path*, i.e. the one connecting the event node with v_{subj} , and *right branch path*, i.e. the one connecting the event node with v_{obj} . For example, in Figure 4 the path P connecting `fred:John_Mccarthy` with `fred:Lisp` includes an event, i.e. `fred:invent_1`. Hence P is a tree, which root is this event node. The left branch path of P is the one connecting `fred:invent_1` with `fred:Lisp`, while the right branch path of P is the one connecting `fred:invent_1` with `fred:John_Mccarthy`. In order to define a property chain axiom OKE needs to define the inverses of all properties in the left branch of P . However, these branch paths may contain properties defined by VerbNet, i.e. thematic roles, which are independent of the event they are associated with,

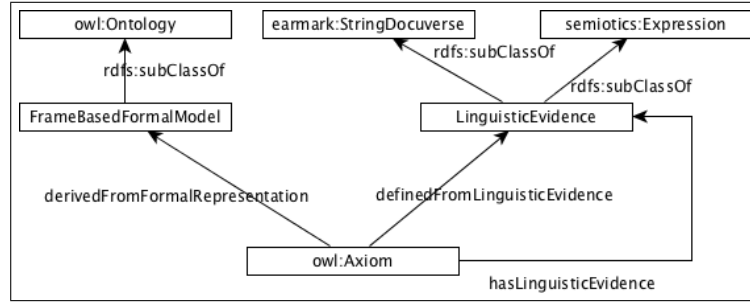


Fig. 6. The grounding vocabulary used for annotating the generated triples and properties with information about their linguistic and formal representation scope.

in the scope of G , i.e. they are general domain properties. In other words, these properties do not carry any information about the event included in the path, which is relevant as far as the formal semantics of the OKE generated property is concerned. OKE tackles this issue by defining a local thematic role property for each VerbNet role participating in the event included in the path. For example, let us consider the property `legalo:inventProgrammingLanguage` in Figure 5. Its reference path includes the two (thematic roles) properties `vn.role:Agent` and `vn.role:Product`. OKE generates two new properties, `legalo:AgentInvent` and `legalo:ProductInvent`, defined as sub-properties of `vn.role:Agent` and `vn.role:Product`, respectively. Given these two new properties, the axioms produced for formalising the generated property `legalo:inventProgrammingLanguage` are given in Example 2.5

Example 2.5. (Property Chain Axiom when the connecting path includes an event.)

```
legalo:inventProgrammingLanguage
  a owl:ObjectProperty ;
  owl:propertyChainAxiom _:b1 .
legalo:AgentInvent
  rdfs:subPropertyOf vn.role:Agent .
legalo:ProductInvent
  rdfs:subPropertyOf vn.role:Product .
_:b1:-( [owl:inverseOf legalo:AgentInvent]
  legalo:ProductInvent) .
```

Scope annotations. Finally, OKE annotates all generated properties and triples with information related to the linguistic and formal representation scopes from which they were derived. To this aim a specific OWL ontology has been defined, named *ground-*

*ing*²⁴, depicted in Figure 6. This ontology reuses Earmark²⁵, a vocabulary for annotating textual content, and *semiotics*²⁶, a content ontology pattern that encodes a basic semiotic theory. Earmark defines the class `earmark:StringDocuverse`, which represents any container of strings that may appear in a document. In the context of OKE this class can be used for representing the sentence s . The semiotics content pattern defines three main classes: Expression, Meaning, Reference (the semiotic triangle). The class Expression is also reused for representing the sentence s . As for the annotation of the linguistic scope of a RDF triple, the grounding vocabulary defines the more specific concept of “linguistic evidence”. In fact, according to the axioms defined in Section 2.2 and the generative rules defined in Section 2.3, the sentence s provides an evidence of the relation $\varphi_s(e_{subj}, e_{obj})$, which is formalised by a RDF triple $(v_{subj}, p_\lambda, v_{obj})$. The concept of “linguistic evidence” is represented by the class `LinguisticEvidence` that specialises both `earmark:StringDocuverse` and `semiotics:Expression`. The OWL property that relates a RDF triple generated by OKE and its linguistic evidence is `hasLinguisticEvidence`.

Additionally, the class `FrameBasedFormalModel` is defined for representing the concept of *frame-based formal representation* of a textual sentence, described in detail in Section 2.1. This class is instantiated by the graph G representing s , which provides the formal scope for all generated properties and triples. The property `derivedFromFormalRepresentation` of the grounding ontology, connects a OKE generated

²⁴The vocabulary can be downloaded from <http://ontologydesignpatterns.org/cp/owl/grounding.owl>

²⁵<http://www.essepuntato.it/2008/12/earmark>

²⁶<http://www.ontologydesignpatterns.org/cp/owl/semiotics.owl>

property as well as a RDF triple, with the graph G from which they were derived. As an example, let us consider the sentence represented by the graph in Figure 4 and the generated RDF triple of the property `legalo:inventProgrammingLanguage` depicted in Figure 5. The scope annotations shown in Example 2.6 are generated.

Example 2.6. (Scope annotations of a OKE property)

```
legalo:sentence
  a grounding:LinguisticEvidence ;
  earmark:hasContent "The New York Times
reported that John McCarthy died. He invented
the programming language LISP." .
[] a owl:Axiom ;
  grounding:hasLinguisticEvidence legalo:sentence;
  owl:annotatedProperty
    legalo:inventProgrammingLanguage ;
  owl:annotatedSource
    dbpedia:John_McCarthy_(computer_scientist);
  owl:annotatedTarget
    dbpedia:Lisp_(programming_language) .
legalo:inventProgrammingLanguage
  a owl:ObjectProperty ;
  grounding:derivedFromFormalRepresentation
    krgraph:52f88ca22 ;
  grounding:definedFromLinguisticEvidence
    legalo:sentence .
```

The first two axioms simply create an individual of type `LinguisticEvidence` for representing the sentence. The second group of axioms annotates the RDF triple for “John McCarthy invented Lisp” with its linguistic evidence.

Finally, the `legalo:inventProgrammingLanguage` property is annotated with its linguistic as well as its formal scope.

2.5. Alignment to Semantic Web vocabularies

This step has the goal of aligning the generated properties to existing Semantic Web ones. The idea is to maximise reuse and linking of extracted knowledge to existing Linked Data. OKE method does not define a specific procedure for addressing this task, rather this step is included in the method for emphasising its importance. The current version of OKE prototype, named *Legalo* (cf. Section 4) implements a simple string matching technique based on the Levenshtein distance measure. The implementation of more sophisticated approaches for aligning OKE properties

to existing vocabularies is part of future work. Relevant related work are ontology matching techniques such as [12] (cf. see the Ontology Alignment Evaluation Initiative²⁷). A possible strategy is to apply state-of-the-art techniques in ontology matching exploiting the information and features provided by the formalisation step (cf. Section 2.4).

3. Data sources

In the context of this work a number of data sources were used for different purposes.

3.1. Abstractive property design

- **WiBi** [15] is a Wikipedia bitaxonomy: a refined, rich and high quality taxonomy that integrates Wikipedia pages and categories. WiBi is used as a reference semantic resource for property label design. WiBi types are exploited in OKE “paraphrasing” task (i.e. abstractive step) when the extracted terms are too general to be informative enough. Details on how WiBi types are used are given in Section 2.3.
- **VerbNet** [44] is the largest domain-independent hierarchical verb lexicon, available for English. It is organized into verb classes. Each verb class is described by thematic roles, selectional restrictions on the arguments, and frames. We map a subset of VerbNet thematic roles to specific prepositions, which are used in OKE paraphrasing task. Details on this mapping are given in Section 2.3.

3.2. Properties alignment to existing Semantic Web vocabularies

Given a sentence including a link, OKE creates a RDF property synthesising the link’s semantics. When possible, it is desirable to align this property to existing Semantic Web properties. The current implementation of OKE uses three semantic resources for addressing this task:

- **Watson**²⁸ [10] is a service that provides access to Semantic Web knowledge, in particular ontologies;

²⁷<http://oaei.ontologymatching.org/>

²⁸<http://watson.kmi.open.ac.uk/WatsonWUI/>

- **Linked Open Vocabularies (LOV)**²⁹ is an aggregator of Linked Open vocabularies (including DBpedia), and provides services for accessing their data;
- **Never-Ending Language Learning (NELL)**³⁰ [6] is a machine learning system that extracts structured data from unstructured Web pages and stores it in a knowledge base. It runs continuously since 2010. From the learnt facts, NELL team has derived an ontology of categories and properties: it includes 548 properties at the moment³¹.

3.3. Evaluation

- **Relation extraction corpus** In order to evaluate Legalo, i.e. the current implementation of OKE, a corpus for relation extraction developed at google research³² was used. Each dataset composing the corpus, contains Wikipedia snippets evaluated by at least five raters to assess if the snippet provides an evidence for a specific relation. Section 5 provides additional details on how this corpus was used in the context of this work;
- **Wikipedia and Wikipedia Pagelinks.** Wikipedia is a collaboratively built encyclopedia on the Web. Currently, English Wikipedia contains ~4.6M articles³³. Each Wikipedia page refers to one entity: these entities are represented in DBpedia, the RDF version of Wikipedia. The Wikipedia Pagelinks dataset³⁴ represents internal links between DBpedia instances as they occur in their corresponding Wikipedia pages. This dataset counts ~149.7M `dbpo:wikiPageWikiLink` triples (as of DBpedia 2014). A subset of Wikipedia pages and their pagelinks³⁵ as testing sample for evaluating the specialised version of OKE implementation for typing Wikipedia pagelinks.

²⁹<http://lov.okfn.org/dataset/lov/>

³⁰<http://rtw.ml.cmu.edu/rtw/>

³¹<http://nelli-ld.telecom-st-etienne.fr/>

³²<https://code.google.com/p/relation-extraction-corpus/downloads/list>

³³Source: http://en.wikipedia.org/wiki/Wikipedia:Size_of_Wikipedi, November 2014

³⁴<http://wiki.dbpedia.org/Downloads39>

³⁵The previous version of the Wikipedia Pagelink dataset (DBpedia 3.9) was used in the evaluation, which counted ~136.6M triples, as the evaluation was performed before the release of DBpedia 2014.

4. Legalo prototype

In this section, an implementation of OKE, named *Legalo*, is presented³⁶. It is based on a pipeline of components and data sources, executed in the sequence illustrated in Figure 7.

1. FRED: Semantic Web machine reader. The core component of the system is *FRED* [41], a Semantic Web machine reader able to produce a RDF/OWL frame-based representation of a text. It integrates, transforms, improves, and abstracts the output of several NLP tools. It performs deep semantic parsing by reusing Boxer [4], which in turn uses a statistical parser (C&C) producing Combinatory Categorical Grammar trees, and thousands of heuristics that exploit existing lexical resources and gazetteers to generate structures according to Discourse Representation Theory (DRT) [25]. The basic NLP tasks performed by Boxer, and reused by FRED, include: event detection, semantic role labeling, first-order logic representation of predicate-argument structures, logical operators scoping (called *boxing*), modality detection, and tense representation.

FRED reengineers DRT/Boxing discourse representation structures according to SW and linked data design practices in order to represent events, role labeling, and boxing as typed n -ary logical patterns in RDF/OWL. The main class for typing events in FRED is `dul:Event`³⁷. In addition, some variables created by Boxer as discourse referents are reified as individuals when they refer to something that has a role in the formal semantics of the sentence. For example, `cat_1` is the reification of the x variable in the first-order predication $Cat(x)$ extracted from the sentence *The cat is on the mat*.

Linguistic Frames [34], Ontology Design Patterns [20], open data, and various vocabularies are reused throughout FRED's pipeline in order to resolve, align, or enrich extracted data and ontologies. The most used include: VerbNet³⁸, for disambiguation of verb-based events; WordNet-RDF³⁹ and OntoWordNet [19] for the alignment of classes to WordNet and DOLCE; DBpedia for the resolution and/or disambiguation of

³⁶A demo of Legalo is available at <http://wit.istc.cnr.it/stlab-tools/legalo/>

³⁷Prefix `dul:` stands for <http://www.ontologydesignpatterns.org/ont/dul/dul.owl#>

³⁸<http://verbs.colorado.edu/~mpalmer/projects/verbnet.html>

³⁹<http://www.w3.org/TR/wordnet-rdf/>

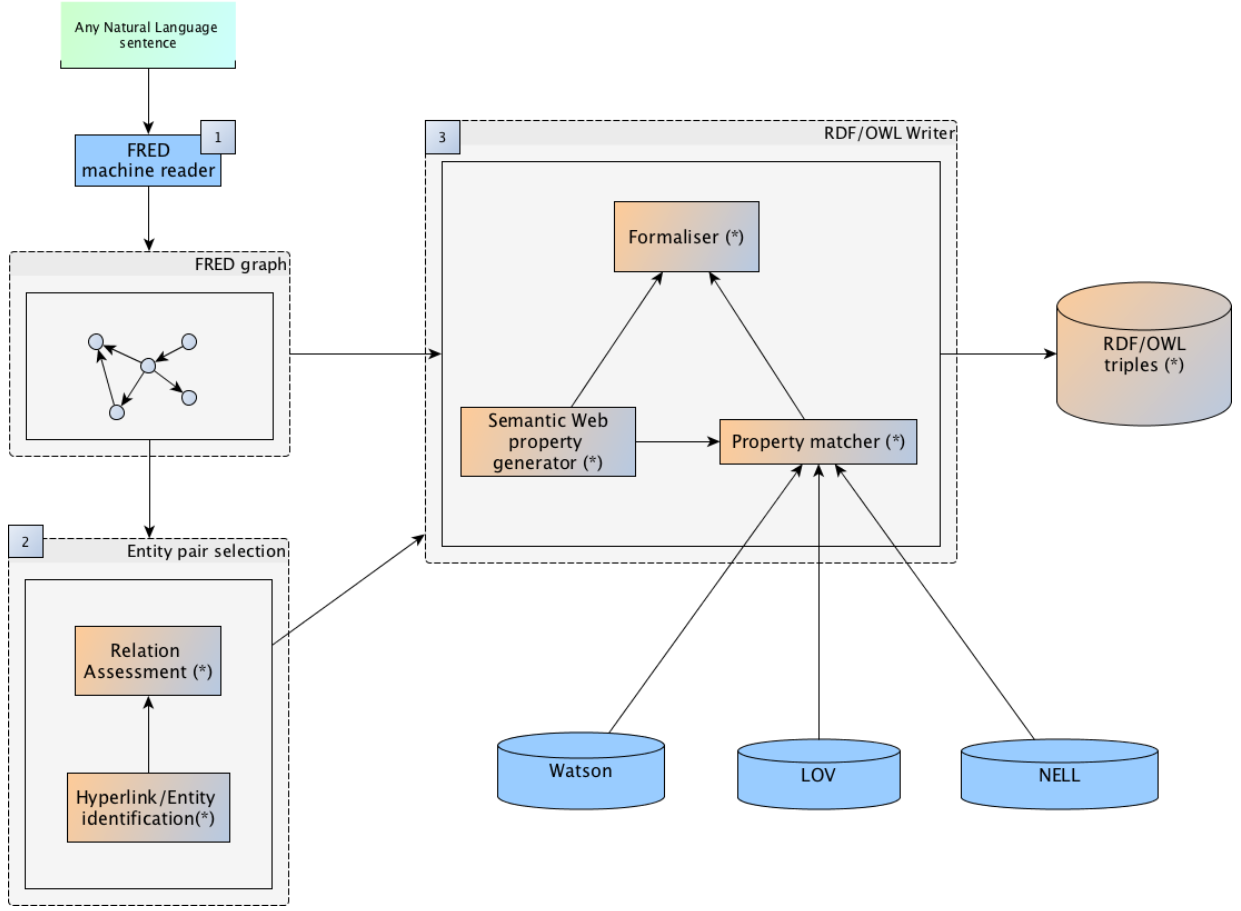


Fig. 7. Pipeline implemented by Legalo for generating Semantic Web properties for semantic annotation of hyperlinks based on their linguistic trace, i.e. natural language sentence including the hyperlinks. Numbers indicate the order of execution of a component in the pipeline. Edges indicates input/output flows. (*) denotes tools developed in this work, which are part this paper contribution.

named entities, as well as for enriching the graph with existing facts known to hold between those entities; schema.org (among others) for typing the recognized named entities. For Named Entity Recognition (NER) and Resolution (a.k.a. Entity Linking) FRED relies on TAGME [13], an algorithmic NER resolver to Wikipedia that heavily uses sentence and Wikipedia context to disambiguate named entities.

All figures depicted in Section 2 show examples of FRED outputs: the reader may want to consider Figure 3, which show the RDF/OWL graph for the sentence “In February 2009 Evile began the pre-production process for their second album with Russ Russell” as a representative output of FRED.

Additionally, FRED reuses The Earmark vocabulary and annotation method [37] for annotating text seg-

ments with the resources from its graphs⁴⁰. For example, in the example sentence of Figure 3, the term “Evil”, starting from the text span “17” and ending at the text span “22” denotes the entity `fred:Evil` in the FRED graph G . This information is formalised with the following triples⁴¹:

```
fred:offset_17_22_Evil
  a pos:PointerRange ;
  rdfs:label "Evil"^^xmls:string ;
  semio:denotes fred:Australia ;
```

⁴⁰These triples are not returned in the graph-view result of FRED at <http://wit.istc.cnr.it/stlab-tools/fred/>, they are returned with all other serialization output options.

⁴¹Prefix `pos:` stands for <http://www.essepuntato.it/2008/12/earmark#>, `semio:` stands for <http://ontologydesignpatterns.org/cp/owl/semiotics.owl#>, and `xmls:` stands for <http://www.w3.org/2001/XMLSchema#>

```
pos:begins "17"^^xmls:nonNegativeInteger ;
pos:ends "22"^^xmls:nonNegativeInteger ;
```

2. Entity pair selection. This component is in charge of detecting the resolved entities and associate them with their lexical surface in s . This is done by querying FRED text span annotations. Another task of this component is, for each pair of detected entities (v_{subj}, v_{obj}) , to assess the existence of φ_s between them. In other words, this component checks the existence paths between v_{subj} and v_{obj} (cf. Axiom 1), selects the shortest one and verifies if there are event nodes in the selected path. If so, it verifies if v_{subj} participates in the event occurrence with an agentive role (cf. Axiom 3). All selected pairs and associated paths are passed to the next component.

3. RDF/OWL writer. This component is in charge of generating a predicate for each pair of entities received in input from the previous component, by applying the generative rules described in Section 2.3 to its associated path. In addition, this components implements two more modules: the “Property matcher” and the “Formaliser”.

The “Property matcher” is in charge of finding alignments between the generated predicate, and existing Semantic Web vocabularies. As described in Section 3, three main sources are used for retrieving semantic property candidates. For assessing their similarity with the generated predicate a string matching algorithm was implemented, which computes a Levenshtein distance metrics [33] between the IDs of the two predicates. Of course, this component is not intended to be a contribution to advance the state of the art in ontology matching, its goal is to contribute to a complete implementation of OKE and to provide a possible baseline for comparing results with future improved versions.

Finally, the RDF/OWL writer includes the component “Formaliser”. This component implements the formalisation step of the method (cf. Section 2.4). It is in charge of producing the triples summarising the relation expressed in s , and that can be used for annotating the corresponding hyperlink, to generate OWL axioms defining domain and range of the generated predicates, and finally to annotate the produced triples and predicates with scope information.

Legalo for typing Wikipedia pagelinks. A specialised version of Legalo for typing Wikipedia pagelinks

(Legalo-Wikipedia)⁴² was presented in [40]. It depends on Legalo has core component and specialise it with two additional features: (i) a sentence extractor specialised for Wikipedia, and (ii) a subject resolver specialised for Wikipedia. A detailed description of this implementation can be found in [40]. Briefly, Legalo Wikipedia takes in input a DBpedia entity, and retrieves all its pagelinks triples (from the Pagelinks DBpedia dataset). For each pagelink triple it extracts all Wikipedia snippets containing an hyperlink corresponding to the triple (sentence extractor). It then select all the the snippets that contain a lexicalisation of the Wikipedia page subject (subject resolver), by relying on the DBpedia Lexicalizations Dataset⁴³. For example, the wikipage `wp:Ron_Cobb` includes a link to `wp:Sydney` in the sentence:

“In 1972, Cobb moved to [Sydney, Australia](#), where his work appeared in alternative magazines such as [The Digger](#).”

This sentence will be selected and stored as it contains the term “Cobb”, which is a lexicalization of `dbpedia:Ron_Cobb`. The same wikipage includes a link to `wp:Los_Angeles_Free_Press` in the sentence:

“Edited and published by [Art Kunkin](#), the [Los Angeles Free Press](#) was one of the first of the underground newspapers of the 1960s, noted for its radical politics.”

This sentence will be discarded as it does not include any lexicalization of `dbpedia:Ron_Cobb`. This procedure is needed for identifying pagelinks that actually convey a semantic factual relation between the Wikipedia page subject and the target of the pagelink. Once the snippets are identified they are passed to Legalo as input together with their associated entity pair.

5. Results and evaluation

The specialised version of Legalo for typing Wikipedia pagelinks has been previously evaluated. For the sake of completeness, these results are summarised in Section 5.2 (for additional details, the reader can refer to [40]). With the help of crowdsourcing an additional,

⁴²A demo is available at <http://wit.istc.cnr.it/stlab-tools/legalo/wikipedia>

⁴³<http://wiki.dbpedia.org/Datasets/NLP?v=yqj>

more extensive evaluation of the current implementation of Legalo was performed, which allowed to better assess its performances and open issues. This section reports this evaluation results in terms of precision, recall, and accuracy.

5.1. Legalo working hypothesis

Legalo is based on two working hypotheses.

Hypothesis 1 (Relevant relation assessment). *Legalo is able to assess if, given a sentence s , a relevant relation exists which holds between two entities, according to the content of s :*

$$\exists \varphi. \varphi_s(e_{subj}, e_{obj})$$

This means that if s contains evidence of a relevant relation between e_{subj} and e_{obj} , then Legalo returns a true value, otherwise it returns false.

Hypothesis 2 (Usable predicate generation). *Legalo is able to generate a usable predicate λ' for a relevant relation φ_s between to entities, expressed in a sentence s : given λ' , a label generated by Legalo for φ_s , and λ_i a label generated by a human for φ_s the following holds (cf. Definition 1):*

$$\lambda' \cong \lambda_i, \lambda_i \in \Lambda$$

which means that the label λ' generated by Legalo is equal or very similar to a label λ_i that a human would define in a Linked Data vocabulary for representing φ_s in a particular textual occurrence.

This section reports the evaluation of Legalo based on the validation of Hypothesis 1 and Hypothesis 2.

Evaluation sample. As evaluation data, a corpus $C_{rel-extraction}$ for relation extraction developed at google research⁴⁴ was used. The available datasets in this corpus are five, and each dataset is dedicated to a specific relation: place of birth, attending or graduating from an institution, place of death, date of death, degree of education. Each dataset includes a snippet from Wikipedia, a pair (subject, object) of freebase entities, and at least five user judgments that indicate if the snippet contains a sentence providing evidence of a referenced relation (e.g., place of death) between the given pair of entities. It is important to remark that Wikipedia snippets included in the corpus contain more than one

sentence, which can be evidence of other relations than the ones for which they were evaluated. Based on this observation, the corpus has been used also for evaluating Legalo on its ability to assess the existence of non-pre-defined relations.

The evaluation was performed using a subset of $C_{rel-extraction}$. More specifically, three evaluation datasets were derived from $C_{rel-extraction}$ and used for performing different experimental tasks.

- $C_{institution}$: a sample of 130 randomly selected snippets extracted from the file of $C_{rel-extraction}$ dedicated to evidence of relations expressing “attending or graduating from an institution”. Legalo was executed on all 130 snippets, including in its input the pair of freebase entities associated with the snippet in $C_{institution}$. For each snippet Legalo gave always an output, either one or more predicates or “no relation evidence” (i.e. false value);
- $C_{education}$: a sample of 130 randomly selected snippets extracted from the file of $C_{rel-extraction}$ dedicated to evidence of relations expressing “obtaining a degree of education”. Legalo was executed on all 130 snippets, including in its input the pair of freebase entities associated with the snippet in $C_{education}$. For each snippet Legalo gave always an output, either one or more predicates or “no relation evidence”;
- $C_{general}$: a sample of 60 randomly selected snippets extracted from $C_{rel-extraction}$, 15 snippets from each file (excluding “date of death” as Legalo only deals with object properties for the moment). The snippets were broken into single sentences and pre-processed with Tagme [13]⁴⁵ in order to enrich them with hyperlinks referring to Wikipedia pages (i.e. DBpedia entities): 186 sentences with at least two recognised DBpedia entities were derived. In total, Legalo produced 867 results, of which 262 predicates and 605 “no relation evidence”. Notice that the high number of false values is not surprising as in many cases a single sentence may contain a high number of entities, and Legalo had to assess the existence of φ on all possible combinations of pairs.

⁴⁴<https://code.google.com/p/relation-extraction-corpus/downloads/list>

⁴⁵<http://tagme.di.unipi.it/>

The resulting triples, predicate formalisations, and scope annotations are accessible via a Virtuoso SPARQL endpoint⁴⁶.

There are several works demonstrating that crowdsourcing can be successfully used for building and evaluating semantic resources [16,47,36]. Following these experiences, Legalo was evaluated with the help of crowdsourcing. Five different types of crowdsourced tasks were defined:

1. assessing if a sentence s is an evidence of the referenced relation (i.e. either “institution” or “education”) between two given entities e_{subj} and e_{obj} mentioned in s - based on data from $C_{institution}$ and $C_{education}$, respectively;
2. assessing if a sentence s is an evidence of any relation between two given entities e_{subj} and e_{obj} mentioned in s - based on data from $C_{general}$;
3. judging if a predicate λ' generated by a machine adequately expresses (i.e. it is a good summarisation of) a specific relation (i.e. either “institution” or “education”) between two given entities e_{subj} and e_{obj} mentioned in s , according to the content of s - based on data from $C_{institution}$ and $C_{education}$, respectively;
4. judging if a predicate λ' generated by a machine adequately expresses (i.e. is a good summarisation of) any relation expressed by the content of s , between two given entities e_{subj} and e_{obj} mentioned in s - based on data from $C_{general}$;
5. creating a phrase λ that summarises the relation expressed by the content of s , between two given entities e_{subj} and e_{obj} mentioned in s - based on data from $C_{general}$.

Task 1 and 2 were used for validating Hypothesis 1. The results of these two tasks were then combined with those from Tasks 3 and 4, for validating Hypothesis 2. Finally, task 5 was used for comparing the similarity between λ values generated by humans and λ' values generated by Legalo, for validating Hypothesis 2 from a different perspective.

It is important to remark that Task 1 duplicates the information already available in $C_{rel-extraction}$: this choice was driven by the need of using smaller datasets ($C_{rel-extraction}$ samples) as Legalo evaluation exper-

iments needed to address different evaluation tasks⁴⁷. Hence, the corpus samples were re-evaluated on the evidence task, in order to ensure a high reliability of the judgements⁴⁸.

The Crowdfunder platform⁴⁹ was used for conducting the crowdsourcing experiments. All tasks included a set of “gold questions” used for computing a trust score t for each worker. Workers had to first perform their job on 7 test questions, and only those reaching $t > 0.7$ were allowed to continue⁵⁰. Given the strong subjective nature of task 5, only for this task a lower trust score $t > 0.6$ was considered acceptable. Each run of a job for a worker contained 4 questions, and they were free to stop contributing at any time. Each question was performed by at least three workers, in order to allow the computation of inter-rater agreement. Besides the initial test questions, in order to keep monitoring workers’ reliability, each job contained one test questions. Results from test questions were excluded from the computation of performance measures (i.e., precision, recall, accuracy, agreement).

For tasks 1 and 2, judgements were expressed as “yes” or “no” answers. For tasks 3 and 4, judgments could be assessed on a scale of three values: Agree (corresponding to a value 1 when computing relevance measures), Partly Agree (corresponding to a value 0.5 when computing relevance measures), and Disagree (corresponding to a value 0 when computing relevant measures). Task 5 was completely open. The *confidence* measure is provided by CrowdFlower, it measures the inter-rater agreement between workers weighted by their trust values, hence indicating both agreement and quality of judgements at the same time. It is computed as described in Definition 5⁵¹, and an example is given in Example 5.1:

Definition 5. (Confidence score)

Given a task unit u , a set of possible judgements $\{j_i\}$, with $i = 1, \dots, n$, a set of trust scores each represent-

⁴⁷Using the whole $C_{rel-extraction}$ corpus for the different tasks was not feasible for the limited time available as well as for the cost of a possible crowdsourcing experiment on such a huge amount of data.

⁴⁸From a preliminary analysis of $C_{rel-extraction}$ it has been noticed that some judgements were incorrect, which can be irrelevant on big numbers while it can bias the results on smaller sets.

⁴⁹<http://www.crowdfunder.com/>

⁵⁰The value range of t is $[0, 1]$, the higher the score, the more reliable the worker.

⁵¹<http://success.crowdfunder.com/customer/portal/articles/1295977>

⁴⁶Legalo results can be inspected at <http://wit.istc.cnr.it:8894/sparql>. The reader can submit a pre-defined default query for retrieving an overview of the dataset.

ing a rater $\{t_k\}$, with $k = 1, \dots, m$, $t_{sum} = \sum_{k=1}^m t_k$ the sum of trust scores of raters giving judgements on u , and $trust(j_i)$ the sum of t_k values of raters that choose judgement j_i , the confidence score $confidence(j_i, u)$ for judgement j_i on the task unit u is computed as follows:

$$confidence(j_i, u) = \frac{trust(j_i)}{t_{sum}}$$

Example 5.1. (Confidence score for evidence judgement)

Table 2 shows the judgements of three raters on the same task unit, where possible judgements are “yes” and “no”. $t_{sum} = 0.95 + 0.89 + 0.98 = 2.82$

Task unit	Judgement	t
582275117	yes	0.95
582275117	no	0.89
582275117	yes	0.98

Table 2

Example of confidence score computation for a task unit.

$$confidence(\text{“yes”}, 582275117) = \frac{0.95+0.98}{2.82} = 0.68$$

$$confidence(\text{“no”}, 582275117) = \frac{0.89}{2.82} = 0.31$$

When aggregating results for a task unit, the judgement with the higher confidence score is selected. Notice that $confidence(j_i, u) = 1$ when all raters give the same judgement.

Evaluation of Hypothesis 1 Table 3 shows the results of the evaluation of Hypothesis 1, i.e. Legalo ability to assess if a sentence s provides evidence of a relation φ_s between two entities (e_{subj}, e_{obj}) . Task 1 was designed for evaluating this capability on specific relations, while Task 2 was designed for evaluating this capability on any relation. Each row shows the performance results for a specific run of the task indicating the type of relation tackled and the crowdsourced task. The most important and informative measure in this case is “precision”. In fact, the value of recall is always 1.0 and accuracy always equals precision. This happens because Legalo always provides either a true or false value and raters can only answer “yes” or “no”. In other words, for this task there can not be neither true negatives or false negatives.

Evaluation of Hypothesis 2 Table 4 shows the results of the evaluation of Hypothesis 2, i.e. Legalo ability of generating usable predicates for summarising relations between entities, according to the content of a sentence. Task 3 was designed for evaluating this capability on specific properties, while Task 4 was designed for evaluating this capability on any property. Each row show the performance results indicating the type of relation tackled and the crowdsourced task. The results for “institution” relation and for “any” relation are computed both on the overall set of results, then on a subset that ensured a higher confidence rate (i.e., only results with $confidence(j_i, u) > 0.65$ are included). As far as the evaluation of the “institution” relation, the subset of results with high confidence is 68% of the whole evaluation dataset, while for “any” relation it is 76%.

Finally, Hypothesis 2 was evaluated also by computing a similarity score between human created predicates and Legalo generated ones. Task 5 was performed for collecting at least three labels λ_i for each triple (s, e_{subj}, e_{obj}) . Surprisingly, the average confidence value on this task was not that low (0.59). We compared Legalo predicate λ' for a triple (s, e_{subj}, e_{obj}) with all λ_i created by the users for that triple. Two different similarity measures were computed: a string similarity score based on Jaccard distance measure⁵², and a semantic similarity measure based on the SimLibrary framework [38]⁵³. The latter is a semantic similarity score that extends string similarity with measures exploiting external semantic resources such as WordNet, MeSH or the Gene Ontology. The average Jaccard similarity score between Legalo labels and human ones is 0.63, while the SimLibrary score is 0.80 (the interval value of both scores is $[0, 1]$, the higher the score, the more similar the the two phrases). Before computing the similarity a pre-processing step was performed to the aim of transforming all verbs to their base form and removing all auxiliary verbs from human predicates. The Stanford CoreNLP framework⁵⁴ was used to compute the lemma and POS tag of each term in the phrase. This lemmatisation step was necessary in order to ensure a

⁵²Given two strings s_1 and s_2 , where c_1 and c_2 are the two character sets of s_1 and s_2 , the Jaccard distance $J(s_1, s_2)$ is defined as $J_{sim}(s_1, s_2) = J_{sim}(c_1, c_2) = \frac{|c_1 \cap c_2|}{|c_1 \cup c_2|}$.

⁵³<http://simlibrary.wordpress.com/>

⁵⁴<http://nlp.stanford.edu/software/corenlp.shtml>

Task	Relation	Precision	Recall	F-measure	Accuracy	Confidence
2	Any	0.84	1.0	0.91	0.84	0.82
1	Education	0.87	1.0	0.93	0.87	0.96
1	Institution	0.84	1.0	0.91	0.84	0.94

Table 3

Results of Legalo performance in assessing the evidence of relations between entity pairs in a given sentence s . Performance measures are computed on the judgements collected in Task 1 and 2 based on data from $C_{institution}$, $C_{education}$, and $C_{general}$.

Task	Relation	Precision	Recall	F-measure	Accuracy	Confidence
3	Education	0.92	0.91	0.91	0.85	0.80
3	Institution	0.65	0.91	0.76	0.62	0.59
3 (high confidence only)	Institution	0.74	0.89	0.81	0.68	0.71
4	Any	0.68	0.90	0.78	0.71	0.64
4 (high confidence only)	Any	0.73	0.87	0.80	0.75	0.76

Table 4

Results of Legalo performance in producing a usable label for relations between entity pairs in a given sentence. Performance measures are computed on the judgements collected in Tasks 3 and 4 based on data from $C_{institution}$, $C_{education}$, and $C_{general}$.

fair comparison of labels based on string similarity as currently Legalo uses only base verb forms.

Evaluating the alignment with existing Semantic Web vocabularies. The matching process performed against LOV⁵⁵, NELL⁵⁶ [6], and Watson⁵⁷ [10] returned a number of proposed alignments between predicates generated by Legalo and existing properties in Linked Data vocabularies. In order to accept an alignment and include it in the formalisation of a Legalo property p_{new} , a threshold $d_{min} = 0.70$ on the computed similarity score (i.e., normalised difference percentage based Levenshtein distance⁵⁸) was set, i.e. only alignments between properties having $d > 0.70$ were kept for the evaluation. All alignments satisfying this requirements were included in the formalisation of the properties generated during this study⁵⁹.

The alignment procedure was executed on 629 Legalo properties p_{new} . For 250 p_{new} , it produced at least one alignment to a Semantic Web property p_{sw} with $d > 0.70$. Three raters independently judged on a scale of three values (Agree, Neutral, Disagree) the resulting alignments based on the available metadata of p_{sw} i.e., comments, labels, domain and range. Ta-

ble 5 shows the results of the user-based evaluation of the alignments between p_{new} and p_{sw} . The three raters have independently judged the proposed alignment very accurate (Precision 0.84) with a high inter-rater agreement (Kendall's W 0.76). Although it was not possible to compute recall for this evaluation, the low percentage of proposed alignments (only 40%) and the simple method applied suggest that there is considerable room for improvement. This evaluation and the implemented method are to be considered a baseline for future work on this specific task.

# p_{new} with at least one p_{sw}	Total # of (p_{new} , p_{sw})	Levenshtein threshold	Precision	Kendall's W
250	693	0.7	0.84	0.76

Table 5

Evaluation results on the accuracy of the alignment between p_{new} and p_{sw} .

5.2. Results and evaluation of Legalo applied to Wikipedia pagelinks

A previous study [40] described the evaluation of a specialised implementation of Legalo for dealing with Wikipedia pagelinks, i.e. Legalo-Wikipedia. In this section the results of this evaluation are reported. The specialised version of Legalo for Wikipedia pagelinks is available online⁶⁰. This version of Legalo works un-

⁵⁵<http://lov.okfn.org/dataset/lov/>

⁵⁶<http://rtw.ml.cmu.edu/rtw/>

⁵⁷<http://watson.kmi.open.ac.uk/WatsonWUI/>

⁵⁸<http://bit.ly/1qd45AQ>

⁵⁹All triples, property formalisations, and alignments can be retrieved at <http://wit.istc.cnr.it:8894/sparql>.

⁶⁰<http://wit.istc.cnr.it/stlab-tools/legalo-wikipedia/>

der the assumption that pagelinks in a Wikipedia page correspond to a semantic relation between the subject of the Wikipage containing the link, and the entity referred by the target page of the link. The linguistic trace of a pagelink is extracted from the text surrounding the hyperlink, and for each pair $(s, \text{hyperlink})$ Legalo generates a predicate if a path between the two entities exists in the graph G that formally represents s . The main difference between Legalo and its Wikipedia specialised version is that in the latter, the subject of the predicate is always given and there is a high probability that it is correct based on the design principles that guide Wikipedia page writing. More precisely, Legalo has to tackle with identifying the subject of a relation, which can be any of the entities recognised in s . Furthermore, the evaluation experiment of Legalo-Wikipedia was performed by experts of Linked Data, hence comparing the new results with the previous ones provides insights on the usability of the generated predicates regardless the expertise of the evaluators.

The evaluation results of Legalo-Wikipedia are published as RDF data and accessible through a SPARQL endpoint⁶¹.

The evaluated sample set consisted of 629 pairs $(s, \text{hyperlink})$, each associated with a FRED graph G . Legalo was executed on this corpus and generated 629 predicates (referred to as p_{new} from now on). The user-based evaluation involved three raters, who are computer science researchers familiar with Linked Data, but not familiar with Legalo. Independently, they have judged the results of Legalo based on two separate tasks, using a Likert scale of five values (Strongly Agree, Agree, Partly Agree, Disagree, Strongly Disagree). When computing performance measures the scale was reduced to three values. Specifically, Strongly Disagree and Agree were associated with a value 1, Partly Agree with 0.5, and Disagree and Strongly Disagree with 0.

The results of the user-based evaluation of p_{new} are reported in Table 6. The three raters have independently judged that the generated predicates p_{new} were very well designed and accurate (F-measure 0.83) in capturing the semantics of their associated pagelinks according to the content of the sentence s , with a high inter-rater agreement (Kendall's W 0.73).

Number of p_{new}	Precision	Recall	F-measure	Kendall's W ⁶²
629	0.72	0.97	0.83	0.73

Table 6

Evaluation results on the accuracy of p_{new} .

6. Discussion

Evaluation results. The results of the crowdsourced tasks demonstrate that Legalo method has high performance (average F-measure=0.92) on the assessment of $\varphi_s(v_{subj}, v_{obj})$ existence (cf. Hypothesis 1). These results are really satisfactory especially compared with performance results of Legalo-Wikipedia [40], where this aspect was not tackled, and φ_s existence was partly ensured by the nature of input data (cf. see also Section 5.2).

As far as Hypothesis 2 (i.e. the usability of generated λ') is concerned, Legalo also in this case show very satisfactory performance. An impressive result is the high average value of the semantic similarity score (0.80) between user created predicates and Legalo generated ones. This result confirms the hypothesis discussed in [40], saying that Legalo design strategy was good at producing predicates that are very close to what a human would do when creating a Linked Data vocabulary. In the context of this work, this hypothesis can be extended to the capability to summarise such relations in a way very close to what a generic user would do. This result is very promising from the perspective of evolving Legalo into a summarisation tool, which is one of the envisioned directions of research.

Improving usability of generated predicates Nevertheless, there is room for improvement, and this is better figured by analysing the results on the evaluation of Hypothesis 2, i.e. the usability of generated λ' . Results show that also on this task Legalo has high performance, with an average F-measure=0.82, however, by inspecting the different relevance measures, it emerges that while recall is very high on all tasks (0.90 on average), average accuracy is 0.73 and average precision is 0.75. Although these are very satisfactory performances, it is worth identifying the cases that cause the generation of less usable or even bad results. An insight is that lower precision and accuracy are registered especially in the generation of predicates for “institution” (accuracy 0.62, precision 0.65) relations and for “any” relations (accuracy 0.71, precision 0.68) while for “education” relations these measures show significantly higher values (accuracy 0.85, precision 0.92). This turns out as an important lesson learnt. In fact, less satisfactory precision seems due to the fact that many “institution” relations between two entities

⁶¹<http://isotta.cs.unibo.it:9191/sparql>

(X, Z) are described in the form “X received his Y from institution Z” (or similar), i.e. a ternary relation, which in a frame-based representation G corresponds to something like:

```
:receive_1 vn.role:Agent :X ;
:receive_1 vn.role:Theme :Y ;
:receive_1 vn.role:Source :Z .
```

Currently, based on this representation, Legalo would generate a predicate by following the path connecting X to Y , hence without considering the information on Y . The resulting predicate in this case would be “receive from”, while a more informative and usable one would clearly be, e.g. “receive degree from”, assuming that the type of Y is degree⁶³. This case can be easily generalised by exploiting the semantic information about the thematic role that Y plays in participating in the event *receive_1*. In fact, a representation pattern can be recognised here: when participating in the event *receive_1*, X plays an agentive role (as expected from Axiom 3), Y plays a passive role, and Z plays an oblique role. The type of an entity playing a passive role, i.e. Y in this case, is a relevant information as far as the relation between an entity playing an agentive role, and another playing an oblique role in an event, is concerned. This pattern can be generalised to other relations than institution, which explains a similar behaviour of Legalo in the two tasks focusing on assessing usability of predicates for “institution” and “any” relations.

Another example that shows this pattern is given by the sentence,

“Hassan Hussein became an organizer for the Communist Party.”

taken from the dataset $C_{general}$. In this case, the representation is the following:

```
:become_1 vn.role:Agent Hassan_Husseini ;
:become_1 vn.role:Patient :Organizer ;
:become_1 :for :Communist_Party .
```

and Legalo would produce the predicate “become for”. By applying the new suggested generative rule, the generated predicate would be instead, the more informative and usable “become organizer for”. This type of observations leads to the definition of additional generative rules that refine OKE and its implementation towards a highly probable improve-

⁶³The term degree is an example of a possible type for Y , however whatever is the type of Y including its type in the predicate would make it much more informative and usable.

ment on precision and accuracy. New rules are currently being implemented based on the data collected on the different tasks, hence the reader may find an evolved and better version of Legalo demo⁶⁴ in short time.

Alignment to existing Semantic Web properties. As for the alignment procedure, there is also space for improvement, since this task was addressed by computing a simple Levenshtein distance. More sophisticated alignment methods such as those from the Ontology Alignment Initiative⁶⁵ or other approaches for entity linking such as SILK⁶⁶ [22] can be investigated for enhancing the alignment results. An interesting result is that our alignment results are good in terms of precision, although all properties that have been matched with a distance score > 0.70 came only from Watson⁶⁷ [10] and LOV⁶⁸. We observed that almost all properties retrieved from NELL⁶⁹ [6] had an editing distance < 0.70 hence almost none of them were judged appropriate. This reinforces the hypothesis the OKE generative rules simulate very well the results of human property creation, i.e. property names are cognitively well designed. In fact, Watson and LOV are repositories of Semantic Web authored ontologies and vocabularies, while NELL properties result from an artificial concatenation of categories learnt automatically.

As for the alignment recall, it was not possible to compute a standard recall metrics because it is impossible to compute False Negative results i.e., all existing Semantic Web properties that would match p_{new} but that we did not retrieve. The relatively high number of missing properties suggests on one hand that a more sophisticated alignment method is needed. On the other hand, if we combine this result with the high value of accuracy of p_{new} and the proposed alignments between p_{new} and p_{sw} , it is reasonable to hypothesise that many cases reveal a lack of intensional coverage in Semantic Web vocabularies, and that OKE can help filling this gap.

Comparison to Open Information Extraction Extracting, discovering, or summarizing relations from text is not an easy task. Natural language is very subtle in providing forms that can express, allude, or entail relations, and syntax offers complex solutions to relate explicitly named entities, anaphoras to mentioned or alluded entities, concepts, and entire phrases, let alone tacit knowledge. Table 7 shows some kinds of (formalisable) relations that can be derived from text.

A full-fledged analysis of those texts is possible to a certain extent, specially if associated with background knowledge (as FRED does), but the conciseness and directness of

⁶⁴<http://wit.istc.cnr.it/stlab-tools/legalo/>

⁶⁵<http://oei.ontologymatching.org/>

⁶⁶<http://wifo5-03.informatik.uni-mannheim.de/bizer/silk/>

⁶⁷<http://watson.kmi.open.ac.uk/WatsonWUI/>

⁶⁸<http://lov.okfn.org/dataset/lov/>

⁶⁹<http://rtw.ml.cmu.edu/rtw/>

sentence	argument#1	binary relation	argument#2
Mr. Miller, 25, entered North Korea seven months ago.	Mr_Miller	enter	North_Korea
He was charged with unruly behavior.	Mr_Miller	charge_with	x:unruly_behavior
North Korean officials suspected he was trying to get inside one of the country's feared prison camps.	y:North_Korean_officials	suspect	try(he, (get_inside (he, z:Korea's_feared_prison_camp)))

Table 7

Sample sentences involving non-trivial relations, expressed in a generic logical form.

hyper-linking based on binary relations is often lost. Hence the importance of tools like Legalo, which are able to re-construct binary relations from complex machine reading graphs.

It would be natural to compare the results of Legalo to relation extraction systems, but this would require to manipulate their output, which is beyond the scope of this work. Here follows an explanation of the difficulties involved.

A state-of-art tool like Open Information Extraction (OIE, [28]) applies an extractive approach to relation extraction, and solves the problem by extracting segments that can be assimilated to subjects, predicates, and objects of a triplet. As reported in [18], its accuracy was not very high with the version of OIE implemented as the ReVerb tool, but it has sensibly improved recently. However, the segments that are extracted, though useful, are not always intuitively reusable as formal RDF properties or individuals. Table 8 shows one case of a very complex segment #3, i.e. “with a West angry over Russia’s actions in Ukraine”, which is a phrase to be further analyzed in order to be formalized, and typically leading to multiple triples; and another case of a complex segment #2, i.e. “developed a passion for the native flora of the arid West Darling region identifying”, which is not easily transformable into a RDF property.

The research presented here intends to go beyond text segmentation, by using an abstractive approach that selects paths in RDF graphs in order to generate RDF properties. The difference between the two approaches is striking, and leads to results that are difficult to compare. Table 9 shows two of the examples from Table 8 (the third one has no resolvable entity on the object position), but as they are extracted and formalized by Legalo.

For the reasons described above, this work has not attempted a direct comparison in terms of accuracy between OIE and Legalo: it would have needed the transformation and formalization of OIE text segments into individuals and properties, and arbitrary choices on how to formalize complex segments. At the end, it is not a measure of their outputs that is obtained, but a measure of authors’ ability to redesign OIE’s output. For those interested in attempts to reuse heterogeneous NLP outputs for formal knowledge extraction, see [18].

7. Related Work

The work presented here can be categorised as *formal binary relation discovery and labeling from arbitrary walks in connected fully-labeled multi-digraphs*, which means in practice that it is not just relation extraction (relations are extracted by FRED [41], and Legalo reuses them), but Legalo discovers complex relations that summarise information encoded in several nodes and edges in the graph (RDF graphs are actually connected, fully-labeled multi-digraphs). It considers certain paths along arbitrary directions of edges, aggregating some of the existing labels, and concatenating them in order to provide property names that are typical of Linked Data vocabularies, and finally axiomatizing the properties with domain, range, subproperty, and property chain axioms.

In other words, Legalo tries to answer the following question: what is the relation that links two (possibly distant) entities in a RDF graph?

There is not much that can be directly comparable in the literature, but work from two related fields can be contrasted with what Legalo does: *relation extraction*, and *automatic summarization*.

The closest works in relation extraction include Open Information Extraction (e.g. [28],[32]), relation extraction exploiting Linked Data [46][26], and question answering on linked data [27].

Relation extraction. The main antecedent to Open Information Extraction is probably the 1999 Open Mind Common Sense project [45], which adopted an ante-litteram crowd-sourcing and games-with-a-purpose approach to populate a large informal knowledge base of facts expressed in triplet-based natural language. The crowd was left substantially free to express the subject, predicate, and object of a triplet, but during its evolution, forms started stabilizing, or were learnt by machine learning algorithms. Currently Open Mind is being merged with several other repositories in ConceptNet [21].

Open Information Extraction (aka Machine Reading) as it is currently known in the NLP community performs *bootstrapped* (i.e. started with learning from a small set of seed examples, and then recursively and incrementally applied to a huge corpus, cf. [11]), *open-domain*, and *unsupervised* in-

segment #1	segment #2	segment #3	sentence
Eugene Nickerson	was quarterback of	the football team and captain	<i>At St. Mark's School in Southborough, Massachusetts, Eugene Nickerson was quarterback of the football team and captain of the hockey team.</i>
President Vladimir Putin	faced	with a West angry over Russia's actions in Ukraine	<i>President Vladimir Putin, faced with a West angry over Russia's actions in Ukraine, has been boosting ties to the East.</i>
Florence May Harding	developed a passion for the native flora of the arid West Darling region identifying	plants	<i>Early in life Florence May Harding developed a passion for the native flora of the arid West Darling region, collecting and identifying plants.</i>

Table 8

Some relations extracted by OIE from sample sentences.

rdf:subject	rdf:property	rdf:object	sentence
dbpedia:Eugene_Nickerson	legalo:quarterbackOf	dbpedia:Amercan_Football	<i>At St. Mark's School in Southborough, Massachusetts, Eugene Nickerson was quarterback of the football team and captain of the hockey team.</i>
dbpedia:Vladimir_Putin	legalo:faceWithAngryOverActionLocatedIn	dbpedia:Ukraine	<i>President Vladimir Putin, faced with a West angry over Russia's actions in Ukraine, has been boosting ties to the East.</i>

Table 9

Two sample extractions by Legalo from the same sentences as in Table 8.

formation extraction. E.g. OIE⁷⁰ is based on learning frequent triplet patterns from a huge shallow parsing of the Web, in order to create a huge knowledge base of triplets composed of text chunks.

This idea (on a smaller scale) was explored in [7], with the goal of resolving predicates to, or to enlarge, a biomedical ontology. On the contrary, OIE extracts binary relations by segmenting the texts into triplets. However, there is usually no attempt to resolve the subjects and objects of those triplets, nor to disambiguate or harmonize the predicates used in the triples. Since predicates are not formally represented, they are hardly reusable for e.g. annotating links with RDFa tags. See Section 6 for a comparison between OIE and Legalo, proving the difficulty of even designing a comparison test.

Overall, Open Information Extraction looks like a component for extractive summarization (see below). In [32], named entity resolution is used to resolve the subjects and objects, and there is an attempt to build a taxonomy of predicates, which are encoded as lexico-syntactic patterns rather than typical predicates.

Another important Open Information Extraction project is Never Ending Language Learning (NELL)⁷¹ [6], a learning tool that since 2010 processes the web for building an evolving knowledge base of facts, categories and relations. In this case there is a (shallow) attempt to build a structured ontology of recognized entities and predicates from the facts learnt by NELL. In this work, NELL is used in an attempt to align the semantic relations resulting from Legalo to NELL ontology.

The main difference between approaches such as OIE and NELL, and Legalo is that the former focus on extracting mainly direct relations between entities, while Legalo fo-

cuses on revealing the semantics of relations between entities that can be: a) directly linked, b) implicitly linked, c) suggested by the presence of links in Web pages, d) indirectly linked, i.e. expressed by longer paths or *n*-ary relations. Legalo novelty also resides in performing property label generation. From the acquisition perspective, Legalo is not bootstrapped, but it is open-domain and unsupervised.

Relation extraction and question answering targeted at Linked Data are quite different from both Open Information Extraction and Legalo, since they are oriented at formal knowledge, but they are not bootstrapped, open domain and unsupervised. They typically use a finite vocabulary of predicates (e.g. from DBpedia ontology), and use their extensional interpretation in data (e.g. DBpedia) to either link two entities recognized in some text (as in [46][26]), or to find an answer to a question, from which some entities have been recognized (as in [27]). Domain is therefore limited to the coverage of the vocabulary, and distant supervision is provided by the background knowledge (e.g. [1]). A growing repository of relationships extracted with this specific domain, distantly supervised approach is sar-graphs [46].

Automatic summarisation. Automatic summarization deserves a short discussion, since ultimately Legalo's relation discovery can be used as a component for that application task. According to [42], the main goal of a summary is to present the main ideas from one or more documents in less space, typically less than half of one document. Different categorizations of summaries have been proposed: topic-based, indicative, generic, etc., but the most relevant seems to distinguish between "extracts" and "abstracts". Extracts are summaries created by reusing portions of the input text verbatim, while abstracts are created by reformulating or regenerating the extracted content. An extraction step is needed in any case, but while extracts *compress* the text by squeezing out unimportant material, and *fuse* the reused portions, abstracts typically *model* the text, by accessing external infor-

⁷⁰<http://openie.cs.washington.edu/>

⁷¹<http://rtw.ml.cmu.edu/rtw/>

mation, applying frames, deep parsing, etc., eventually generating a summary that in principle could contain no word in common with the original text.

Extractive summarization is now in mass usage, e.g. with snippets provided by search engines. It has serious limits of course, because size and relevance of the extracts can be questionable and not as accurate as a human may be.

Legalo can be considered closer to abstractive summarization, since it can be used to build frame-based abstractive summaries of texts, consisting in binary relation discovery, which can then be filtered for relevance. The current implementation of Legalo is not designed in view of abstractive summarization, therefore it was not evaluated it for that task, but it is appropriate to report at least one relevant example of related work in this area.

Opinosis [17] is the state-of-the-art system for abstractive summarisation. It performs graph-based summarisation, generating concise abstractive summaries of highly redundant opinions. It uses a word graph data structure to represent the text, whereas Legalo uses a semantic graph. As the authors say: “Opinosis is a shallow abstractive summariser as it uses the original text itself to generate summaries. This is unlike a true abstractive summariser that would need a deeper level of natural language understanding”. Legalo is indeed based on FRED [41], which provides such deeper level of understanding.

In order to be considered an abstractive summariser, Legalo will need to be complemented with more capabilities to rank discovered relations across an entire or even multiple texts, to associate them in a way that final users can make sense of, and to evaluate summaries appropriately. Results from both abstractive summarisation (e.g. [49][17][24]) and RDF graph summary (e.g. [48][39][5]) can be reused to that purpose.

8. Conclusion and future work

Conclusion. This paper presents a novel approach for Open Knowledge Extraction, and its implementation called *Legalo*, for uncovering the semantics of hyperlinks based on frame-based formal representation of natural language text, and heuristics associated with subgraph patterns. The main novel aspects of the approach are: property label generation, automatic link tagging, graph-based relation extraction, abstractive summarisation.

The working hypothesis is that hyperlinks (either created by humans or knowledge extraction tools) provide a pragmatic trace of semantic relations between two entities, and that such semantic relations, their subjects and objects, can be revealed by processing their linguistic traces: the sentences that embed the hyperlinks. Evaluation experiments conducted with the help of a crowdsourcing platform confirm the hypothesis, and show very high performances: the method is able to predict the actual presence of a relation

with a high precision (average F-measure 0.92), and generate accurate RDF properties between the hyperlinked entities in single-relation corpora (average F-measure 0.84), the Wikipedia page link corpus (average F-measure 0.84), as well as in the challenging open domain corpus (average F-measure 0.78). The accuracy remains constant across crowdsourced evaluation, and comparison to (crowdsourced) gold standard for the open domain corpus. We also provide alignments to Semantic Web vocabularies with a precision value of 0.84.

A demo of Legalo Web service is available online⁷², as well as the prototype dedicated to Wikipedia pagelinks⁷³, and the binary properties produced in this study can be accessed by means of a sparql endpoint⁷⁴.

Ongoing work. Current work concentrates on designing and testing new heuristics, as required by evidence emerging from experiments and tests (cf. e.g. Section 6), on identifying new ways of aligning the relations generated by Legalo to existing ontologies, and on discovering regularities in the relation taxonomies that are increasingly discovered.

Future work. The main research line for the future is to apply Legalo to application tasks. An obvious one is a real abstractive summarisation task, both at single-text, and multiple-text level, evaluating the results against state-of-the-art tools. The challenges there include at least: (i) managing multiple (and possibly dynamically evolving) Open Knowledge Extraction graphs, (ii) assessing relevance of discovered relations, and their dependence across a same text, or across multiple texts, and (iii) generating factoid sequences that make sense to a final user of abstractive summaries. Also other applications of Legalo are envisaged, including question answering and textual entailment.

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⁷²<http://wit.istc.cnr.it/stlab-tools/legalo>

⁷³<http://wit.istc.cnr.it/stlab-tools/legalo/wikipedia>

⁷⁴<http://isotta.cs.unibo.it:9191/sparql>

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